Research Project Summary

Title: *Geophysical Investigation of Groundwater Resources in Abagana, Anambra State (Electrical Resistivity Method)*

Author: Chiegboka Damian Ikemsinachi

Institution: Nnamdi Azikiwe University

Overview

This project focuses on the investigation of groundwater potential in Abagana, Anambra State, using geophysical methods. Groundwater is a critical resource for domestic, agricultural, and industrial use, especially in areas where surface water is unreliable. The study applies electrical resistivity techniques to understand subsurface conditions and identify potential aquifer zones for sustainable water supply.

Objective

The main aim of this research is to evaluate groundwater potential and aquifer characteristics in the study area. Specifically, the project seeks to:

- Identify potential groundwater-bearing zones
- Analyze subsurface geological layers
- Determine aquifer depth, thickness, and properties
- Provide a scientific basis for sustainable groundwater development

This work also aligns with global efforts toward sustainable water management and access to clean water.

Methodology

The study was carried out using the electrical resistivity method, specifically the Vertical Electrical Sounding (VES) technique with the Schlumberger array configuration.

- A total of 9 VES stations were established across the study area
- Data was collected using resistivity equipment (ABEM Terrameter)
- Electrode spacing was progressively increased to probe deeper subsurface layers
- Data interpretation was done using curve matching and computer-based iteration techniques

This approach allowed for detailed mapping of subsurface resistivity variations and identification of geological layers.

Key Findings

- The subsurface consists of 4–5 geological layers, including topsoil, clay, sand, sandstone, and water-saturated zones
- The water table depth ranges from about 180m to 200m
- Aquifer thickness varies between 56m and 120m, indicating significant groundwater potential
- Hydraulic conductivity and transmissivity values show variation across locations, with VES 1 having the highest groundwater yield potential
- Some locations showed shallower water tables (~80m), indicating more accessible groundwater zones

These results demonstrate that groundwater resources are present but vary in quality and accessibility across the area.

Significance

This project provides valuable insights into groundwater distribution in Abagana and supports:

- Sustainable water resource management
- Improved decision-making for borehole drilling
- Reduced risk of groundwater overexploitation

It also highlights the importance of geophysical methods in solving real-world environmental and resource challenges.

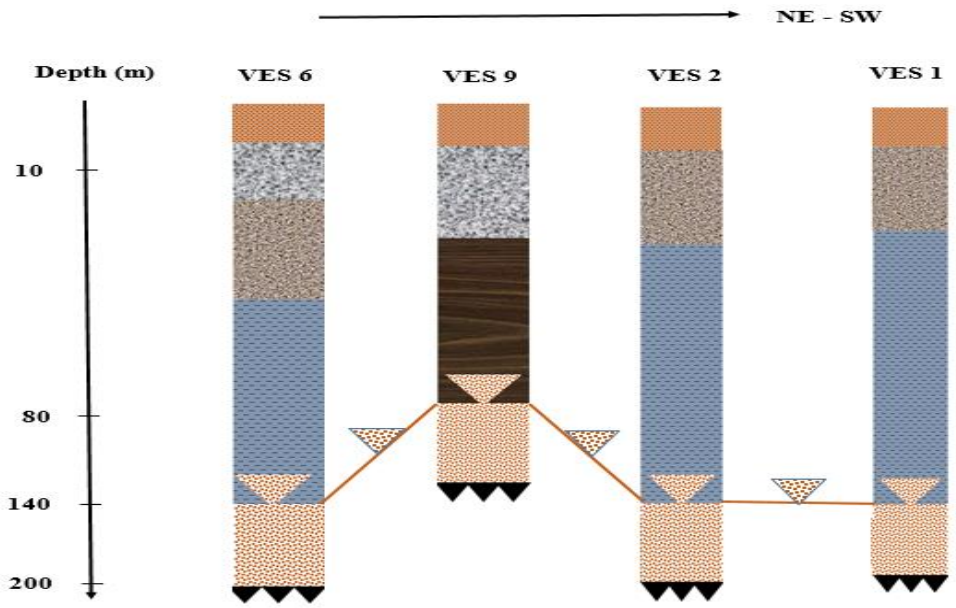

Fig 1: Correlation of Geo-electric layers along North east-South west Direction

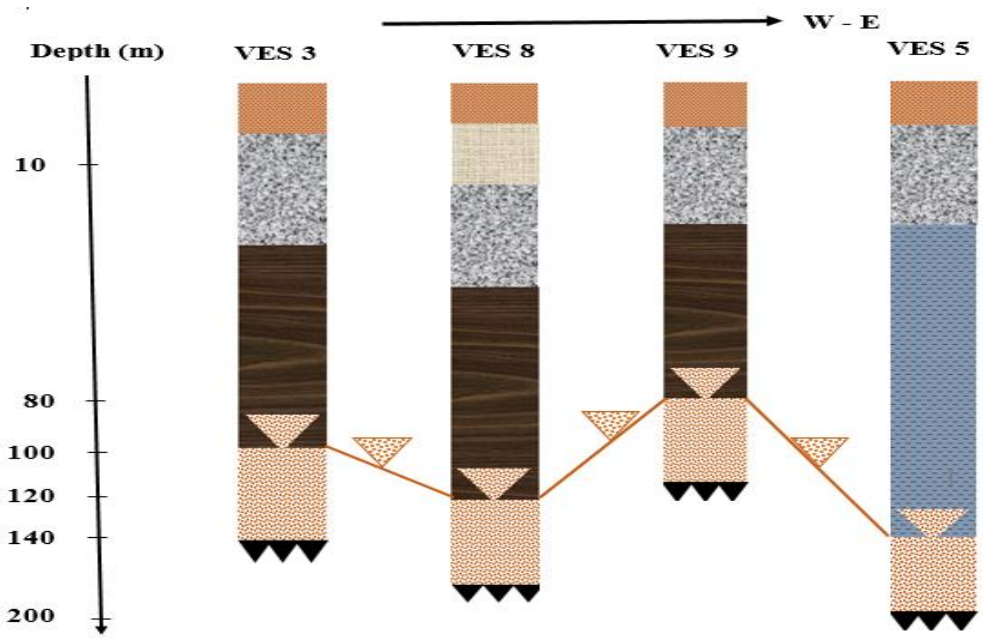

Fig 2 Correlation of Geo-electric layers along West-East Direction

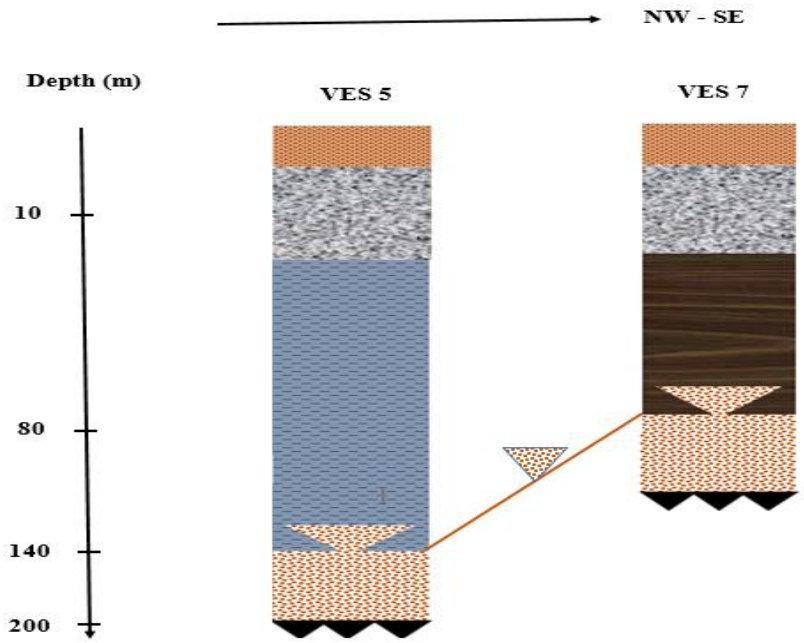

Fig 3: Correlation of Geo-electric layers along North West-South East Direction

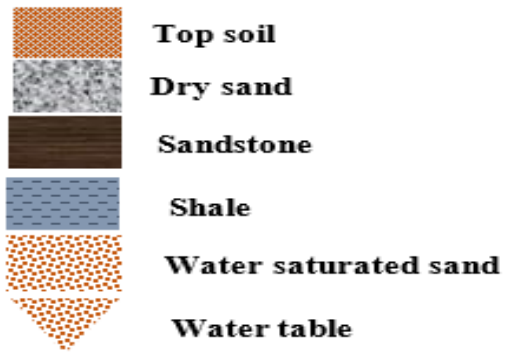

Fig 4: Legend of the Geoelectric Section

Conclusion

The study successfully demonstrates the effectiveness of electrical resistivity methods in groundwater exploration. The findings contribute to a better understanding of subsurface conditions and provide a foundation for future hydrogeological studies and water resource planning in the region.